ODAS Processing-Time Experiment

Run date: 2026-10-04 | **Platform:** Raspberry Pi, ARM64 | **Input:** `sonicwild.cfg`, live `multi_capture`

Method

Four Release builds were run sequentially with the same configuration and physical audio input. An `LD_PRELOAD` probe measured successful, non-empty `mod_ssl_process` and `mod_sst_process` calls with `CLOCK_MONOTONIC`. Values below are average milliseconds per frame.

Version Comparison

Metric	Base `bcb8454` 2024-06-18	YAMNet `c603141` 2026-05-19	SST sidecar `18a8a73 2026-07-20	Current `93a1b81` 2026-09-06
Frames	1,563	1,266	1,563	1,469
SSL average (ms/frame)	0.497371	0.860922	0.890678	0.922171
SST average (ms/frame)	0.012994	0.016728	0.014863	0.088655
SSL + SST / 8 ms frame	6.38%	10.97%	11.32%	12.64%

The share is `(SSL average + SST average) / 8 ms x 100`; it is not end-to-end latency. Frame counts differ, so compare averages rather than total times.

Stage Breakdown: Current Profiler Run

This separate detailed run profiled 1,579 SSL frames and 1,578 SST frames. The current-version stage averages are shown by component; earlier checkpoints do not have this stage-level instrumentation.

SSL stage	Avg (ms/frame)	SST stage	Avg (ms/frame)
Pre-XCorr	0.083	Track Management	0.002
GCC-PHAT XCorr	0.242	Predict	0.001
SRP Accumulation	0.155	Coherence	0.001
Peak Extraction + Mapping	0.002	Mixture + Assignment	0.009
Freq at Peak	0.319	Update	0.001
Format DOA Angles	0.043	Activity + Classify	0.220
Full SSL Frame	**0.843**	Transitions + Dynamic	0.002
		Output Copy	0.001
		Full SST Frame	**0.236**

In this detailed run, SSL and SST together averaged 1.079 ms, or 13.5% of an 8 ms frame interval. Component values are rounded and may not sum exactly to the reported full-frame averages. “Activity + Classify” includes work beyond YAMNet inference and is not an isolated YAMNet measurement. These figures do not represent full-pipeline or worst-case latency.